

Detailed API Contract for Each Component Interaction

Module: Principal Timetable Page

Product: NASSAQ

1) API Design Principles

قبل التفاصيل، يجب الالتزام بهذه القواعد:

1.1 Context Required in Every Request

خاص بالجدول يجب أن يعمل ضمن سياق واضح API كل

- school_id
- academic_year_id
- term_id

إما:

- path params في
- query params أو في
- session/current school context أو يتم استنتاجها من

الأفضل:

عند الحاجة بوضوح body أو query تمريرها في.

1.2 Response Envelope Standard

يفضل توحيد شكل الرد:

```
{  
  
  "success": true,  
  
  "message": "Request completed successfully",  
  
  "data": {},  
}
```

```
"meta": {},  
  
"errors": []  
  
}  
  
وفي حالة الخطأ:  
  
{  
  
  "success": false,  
  
  "message": "Validation failed",  
  
  "data": null,  
  
  "meta": {},  
  
  "errors": [  
  
    {  
  
      "code": "READINESS_BLOCKED",  
  
      "field": "teacher_subject_mapping",  
  
      "message": "Some subjects do not have valid teachers assigned"  
  
    }  
  
  ]  
  
}
```

1.3 Version Safety Rule

يعدل الجدول يجب أن يعمل على API أي:

- Draft version
- جديدة Draft أو ينشئ

مباشر على النسخة المنشورة overwrite ولا يجوز.

2) Page-Level Initial Load APIs

عند التحميل PrincipalTimetablePage التي يحتاجها APIs هذه هي الـ

API 1: Fetch Page Summary

Endpoint

GET /api/principal/timetable/summary

Triggered By

PrincipalTimetablePage.loadPageData()

Purpose

تحميل ملخص عام عن الصفحة:

- school context
- current version state
- counts
- current status banner data

Query Params

```
{  
  
  school_id: string  
  
  academic_year_id: string  
  
  term_id: string  
  
}
```

Response

```
{  
  
  "success": true,  
  
  "message": "Timetable summary loaded successfully",  
  
  "data": {  
  
    "school": {  
  
      "id": "SCH_001",  
  
      "name_ar": "مدرسة النسق النموذجية"  
  
    },  
  
    "academic_year": {  
  
      "id": "AY_2025",  
  
      "name": "2025 - 2026"  
  
    }  
  
  }  
  
}
```

```
},  
  
"term": {  
  
  "id": "TERM_1",  
  
  "name": "الفصل الدراسي الأول"  
  
},  
  
"counts": {  
  
  "classes": 24,  
  
  "teachers": 48,  
  
  "subjects": 16,  
  
  "teaching_slots": 35  
  
},  
  
"current_version": {  
  
  "id": "TV_001",  
  
  "version_name": "Draft v3",  
  
  "status": "draft",  
  
  "quality_score": 87.5,  
  
  "warnings_count": 4,  
  
  "conflicts_count": 1,  
  
  "generated_at": "2026-03-13T10:00:00Z",  
  
  "generated_by": "Ahmad Zalat"  
  
},  
  
"generation_status": "idle"  
  
},  
  
"meta": {},  
  
"errors": []  
  
}
```

Error Cases

- missing active school context
- invalid academic year
- no active term

Used By Components

- PrincipalTimetablePage
- TimetablePageHeader
- TimetableStatusBanner

API 2: Fetch Readiness Summary

Endpoint

GET /api/principal/timetable/readiness

Triggered By

- initial page load
- clicking refresh readiness
- before opening generation modal

Purpose

فحص جاهزية البيانات اللازمة لتوليد الجدول.

Query Params

```
{  
  school_id: string  
  academic_year_id: string  
  term_id: string  
}
```

Response

```
{  
  "success": true,  
  "message": "Readiness summary loaded successfully",  
  "data": {  
    "overall_status": "partial",
```

```
"total_checks": 10,

"passed_checks": 7,

"failed_checks": 2,

"warning_checks": 1,

"items": [

  {

    "key": "school_schedule_settings",

    "label": "إعدادات اليوم الدراسي",

    "status": "complete",

    "description": "تمت تهيئة إعدادات اليوم الدراسي بنجاح",

    "fix_route": null,

    "fix_action_label": null

  },

  {

    "key": "teacher_subject_mapping",

    "label": "ربط المعلمين بالمواد",

    "status": "missing",

    "description": "بعض المواد لا يوجد لها معلم صالح",

    "fix_route": "/principal/users?tab=teachers",

    "fix_action_label": "إصلاح الربط"

  }

]

},

"meta": {},

"errors": []

}
```

Used By Components

- TimetableReadinessPanel
- AITimetableGenerateButton
- AITimetableGenerationModal

API 3: Fetch Versions List

Endpoint

GET /api/principal/timetable/versions

Triggered By

- initial page load
- after generation
- after publish/archive

Query Params

```
{  
  
  school_id: string  
  
  academic_year_id: string  
  
  term_id: string  
  
}
```

Response

```
{  
  
  "success": true,  
  
  "message": "Versions loaded successfully",  
  
  "data": {  
  
    "versions": [  
  
      {  
  
        "id": "TV_001",  
  
        "version_name": "Draft v3",  
  
        "status": "draft",  
  
        "generation_mode": "full",  
  
      }  
    ]  
  }  
}
```

```

    "quality_score": 87.5,

    "conflicts_count": 1,

    "warnings_count": 4,

    "generated_at": "2026-03-13T10:00:00Z",

    "generated_by": "Ahmad Zalat"

  },

  {

    "id": "TV_000",

    "version_name": "Published v2",

    "status": "published",

    "generation_mode": "full",

    "quality_score": 82.1,

    "conflicts_count": 0,

    "warnings_count": 3,

    "generated_at": "2026-03-12T08:00:00Z",

    "generated_by": "System"

  }

]

},

"meta": {},

"errors": []

}

```

Used By Components

- TimetableVersionManager
- TimetableFiltersBar

3) Grid and View Control APIs

API 4: Fetch Timetable Grid

Endpoint

GET /api/principal/timetable/grid

Triggered By

- page load after summary
- changing version
- changing filters
- changing view mode
- refreshing after generation

Purpose

إرجاع البيانات المطلوبة لعرض الجدول.

Query Params

```
{  
  
  school_id: string  
  
  academic_year_id: string  
  
  term_id: string  
  
  version_id: string  
  
  view_mode: "class" | "teacher" | "grade" | "day"  
  
  class_id?: string  
  
  teacher_id?: string  
  
  grade_id?: string  
  
  weekday_number?: number  
  
  subject_id?: string  
  
  search?: string  
  
  show_breaks?: boolean  
  
  show_prayer?: boolean  
  
  core_only?: boolean
```

```
}
```

Response

```
{
```

```
  "success": true,
```

```
  "message": "Timetable grid loaded successfully",
```

```
  "data": {
```

```
    "version": {
```

```
      "id": "TV_001",
```

```
      "status": "draft"
```

```
    },
```

```
    "mode": "class",
```

```
    "columns": [
```

```
      { "key": "day_1", "label": "الأحد" },
```

```
      { "key": "day_2", "label": "الاثنين" },
```

```
      { "key": "day_3", "label": "الثلاثاء" }
```

```
    ],
```

```
    "rows": [
```

```
      {
```

```
        "key": "block_1",
```

```
        "label": "الحصة الأولى",
```

```
        "start_time": "07:30",
```

```
        "end_time": "08:15",
```

```
        "block_type": "period"
```

```
      },
```

```
      {
```

```
        "key": "block_2",
```

```
        "label": "الاستراحة",
```

```
"start_time": "10:00",

"end_time": "10:15",

"block_type": "break"

}

],

"entries": [

{

"id": "TE_001",

"column_key": "day_1",

"row_key": "block_1",

"entry_type": "lecture",

"subject_name": "الرياضيات",

"teacher_name": "أ. محمد علي",

"class_name": "خامس / أ",

"room_name": "R-201",

"is_ai_generated": true,

"is_locked": false,

"has_warning": false,

"warning_count": 0,

"status": "scheduled"

}

]

},

"meta": {

"applied_filters": {

"view_mode": "class",

"class_id": "CLS_12"
```

```
}  
  
},  
  
"errors": []  
  
}
```

Used By Components

- TimetableGridSection
- TimetableGrid
- TimetableCell

API 5: Fetch Filter Options

Endpoint

GET /api/principal/timetable/filter-options

Triggered By

- page load
- when context changes

Purpose

الخاصة بالفلاتر options إرجاع.

Query Params

```
{  
  
  school_id: string  
  
  academic_year_id: string  
  
  term_id: string  
  
}
```

Response

```
{  
  
  "success": true,  
  
  "message": "Filter options loaded successfully",  
  
}
```

```
"data": {  
  
  "classes": [  
  
    { "label": "أ / خامس", "value": "CLS_12" }  
  
  ],  
  
  "teachers": [  
  
    { "label": "أ. محمد علي", "value": "TCH_14" }  
  
  ],  
  
  "grades": [  
  
    { "label": "الصف الخامس", "value": "GR_5" }  
  
  ],  
  
  "subjects": [  
  
    { "label": "الرياضيات", "value": "SUB_8" }  
  
  ],  
  
  "weekdays": [  
  
    { "label": "الأحد", "value": 1 }  
  
  ]  
  
},  
  
"meta": {},  
  
"errors": []  
  
}
```

Used By Components

- TimetableFiltersBar
- PartialRegenerationModal

4) Insights and Issues APIs

API 6: Fetch Insights Summary

Endpoint

GET /api/principal/timetable/insights

Triggered By

- after selecting a version
- after generation completes

Query Params

```
{  
  
  version_id: string  
  
}
```

Response

```
{  
  
  "success": true,  
  
  "message": "Insights loaded successfully",  
  
  "data": {  
  
    "total_required_sessions": 420,  
  
    "total_assigned_sessions": 410,  
  
    "total_unassigned_sessions": 10,  
  
    "quality_score": 87.5,  
  
    "warnings_count": 4,  
  
    "hard_conflicts_count": 1,  
  
    "affected_teachers_count": 3,  
  
    "affected_classes_count": 2,  
  
    "top_overloaded_class": "خامس / أ",  
  
    "top_overloaded_teacher": "أ. محمد علي",  
  
    "top_clustered_subject": "الرياضيات",  
  
    "ai_notes": [  
  
      "يوجد تكدس لمادة الرياضيات في الصف الخامس",  
  
      "بعض الفصول لديها فجوات يمكن تحسينها"
```

```
]
},
"meta": {},
"errors": []
}
```

Used By Components

- TimetableInsightsPanel

API 7: Fetch Issues List

Endpoint

GET /api/principal/timetable/issues

Triggered By

- version selection
- after generation
- after diagnostics refresh

Query Params

```
{
  version_id: string
  severity?: "low" | "medium" | "high" | "critical"
  issue_type?: string
}
```

Response

```
{
  "success": true,
  "message": "Issues loaded successfully",
  "data": {
    "items": [
```

```
{
  "id": "ISS_001",
  "type": "unassigned_unit",
  "severity": "high",
  "title": "حصة غير موزعة",
  "description": "تعذر إسناد حصة علوم لفصل خامس / أ",
  "related_entity_type": "class_subject",
  "related_entity_name": "خامس / أ - علوم",
  "recommended_action": "تحقق من ربط المعلمين أو أعد المعالجة الجزئية"
}
]
},
"meta": {
  "total": 1
},
"errors": []
}
```

Used By Components

- TimetableIssuesSection

5) Diagnostics APIs

API 8: Fetch Diagnostics Summary

Endpoint

GET /api/principal/timetable/diagnostics

Triggered By

- clicking “عرض التشخيص”
- after generation completes if user opens modal

Query Params

```
{  
  version_id: string  
}
```

Response

```
{  
  "success": true,  
  "message": "Diagnostics loaded successfully",  
  "data": {  
    "run_id": "RUN_001",  
    "started_at": "2026-03-13T09:58:00Z",  
    "finished_at": "2026-03-13T10:00:00Z",  
    "status": "completed",  
    "total_units": 420,  
    "assigned_units": 410,  
    "unassigned_units": 10,  
    "hard_conflicts": 1,  
    "soft_conflicts": 4,  
    "logs": [  
      {  
        "group": "readiness",  
        "items": [  
          {  
            "timestamp": "2026-03-13T09:58:02Z",  
            "level": "info",  
            "message": "School settings loaded successfully"  
          }  
        ]  
      }  
    ]  
  }  
}
```

```
]
},
{
  "group": "assignment",
  "items": [
    {
      "timestamp": "2026-03-13T09:59:12Z",
      "level": "warning",
      "message": "Could not assign one science unit for class 5A"
    }
  ]
}
]
}
]
},
"meta": {},
"errors": []
}
```

Used By Components

- TimetableDiagnosticsModal

API 9: Fetch Session Details

Endpoint

GET /api/principal/timetable/session-details

Triggered By

- clicking a cell in the timetable grid

Query Params

```
{
```

```
entry_id: string
}
```

Response

```
{
  "success": true,
  "message": "Session details loaded successfully",
  "data": {
    "id": "TE_001",
    "subject_name": "الرياضيات",
    "teacher_name": "أ. محمد علي",
    "class_name": "خامس / أ",
    "grade_name": "الصف الخامس",
    "weekday_label": "الأحد",
    "start_time": "07:30",
    "end_time": "08:15",
    "entry_type": "lecture",
    "room_name": "R-201",
    "is_ai_generated": true,
    "is_locked": false,
    "status": "scheduled",
    "notes": "تم إسناد الحصة خلال التوليد الكامل",
    "warnings": []
  },
  "meta": {},
  "errors": []
}
```

Used By Components

- TimetableSessionDetailsDrawer

6) Generation APIs

API 10: Generate Full Timetable

Endpoint

POST /api/principal/timetable/generate

Triggered By

- confirming AITimetableGenerationModal

Purpose

بدء عملية توليد جدول كامل.

Request Body

```
{  
  
  "school_id": "SCH_001",  
  
  "academic_year_id": "AY_2025",  
  
  "term_id": "TERM_1",  
  
  "generation_mode": "full",  
  
  "use_published_as_baseline": false  
  
}
```

Success Response

```
{  
  
  "success": true,  
  
  "message": "Timetable generation started successfully",  
  
  "data": {  
  
    "generation_run_id": "RUN_002",  
  
    "version_id": "TV_002",  
  
    "generation_status": "running"  
  
  }  
  
}
```

```
},  
  
"meta": {},  
  
"errors": []  
  
}
```

Alternate Response if synchronous short-running

```
{  
  
  "success": true,  
  
  "message": "Timetable generated successfully",  
  
  "data": {  
  
    "generation_run_id": "RUN_002",  
  
    "version_id": "TV_002",  
  
    "generation_status": "completed",  
  
    "quality_score": 89.4,  
  
    "assigned_units": 417,  
  
    "unassigned_units": 3  
  
  },  
  
  "meta": {},  
  
  "errors": []  
  
}
```

Blocking Error Response

```
{  
  
  "success": false,  
  
  "message": "Readiness validation failed",  
  
  "data": null,  
  
  "meta": {},  
  
  "errors": [  
  
    {
```

```
"code": "READINESS_BLOCKED",  
  
"field": "teacher_subject_mapping",  
  
"message": "Some required subjects are missing teacher assignments"  
  
}  
  
]  
  
}
```

Used By Components

- AITimetableGenerationModal
- PrincipalTimetablePage

API 11: Poll Generation Status

Endpoint

GET /api/principal/timetable/generation-status

Triggered By

- after starting generation
- every few seconds while status is running

Query Params

```
{  
  
  generation_run_id: string  
  
}
```

Response

```
{  
  
  "success": true,  
  
  "message": "Generation status loaded successfully",  
  
  "data": {  
  
    "generation_run_id": "RUN_002",  
  
    "status": "running",  

```

```
"stage": "assignment",

"progress_percentage": 64,

"current_step_label": "توزيع الحصص",

"version_id": "TV_002"

},

"meta": {},

"errors": []

}
```

Final Completed Response

```
{

"success": true,

"message": "Generation completed successfully",

"data": {

"generation_run_id": "RUN_002",

"status": "completed",

"stage": "completed",

"progress_percentage": 100,

"current_step_label": "تم حفظ النسخة",

"version_id": "TV_002",

"quality_score": 89.4,

"assigned_units": 417,

"unassigned_units": 3,

"warnings_count": 2,

"conflicts_count": 0

},

"meta": {},

"errors": []

}
```

```
}
```

Used By Components

- PrincipalTimetablePage
- TimetableStatusBanner

API 12: Partial Regeneration

Endpoint

POST /api/principal/timetable/regenerate-partial

Triggered By

- confirming PartialRegenerationModal

Request Body

```
{  
  "version_id": "TV_002",  
  "scope": "class",  
  "class_id": "CLS_12"  
}
```

Other valid payload examples

```
{  
  "version_id": "TV_002",  
  "scope": "teacher",  
  "teacher_id": "TCH_14"  
}
```

```
{  
  "version_id": "TV_002",  
  "scope": "day",  
  "weekday_number": 2  
}
```

```
{  
  "version_id": "TV_002",  
  "scope": "subject",  
  "subject_id": "SUB_8",  
  "class_id": "CLS_12"  
}
```

Response

```
{  
  "success": true,  
  "message": "Partial regeneration completed successfully",  
  "data": {  
    "generation_run_id": "RUN_003",  
    "version_id": "TV_003",  
    "status": "completed",  
    "scope": "class",  
    "affected_entities": ["CLS_12"],  
    "quality_score": 90.1  
  },  
  "meta": {},  
  "errors": []  
}
```

Used By Components

- PartialRegenerationModal

7) Version Lifecycle APIs

API 13: Publish Draft Version

Endpoint

POST /api/principal/timetable/version/publish

Triggered By

- confirming PublishTimetableVersionModal

Request Body

```
{  
  "version_id": "TV_002"  
}
```

Response

```
{  
  "success": true,  
  "message": "Timetable version published successfully",  
  "data": {  
    "published_version_id": "TV_002",  
    "previous_published_version_id": "TV_001",  
    "status": "published"  
  },  
  "meta": {},  
  "errors": []  
}
```

Failure Example

```
{  
  "success": false,  
  "message": "Cannot publish this version",  
  "data": null,  
  "meta": {},  
  "errors": []  
}
```

```
"errors": [  
  {  
    "code": "BLOCKING_CONFLICTS",  
    "field": "version_id",  
    "message": "This draft contains blocking hard conflicts"  
  }  
]  
}
```

Used By Components

- PublishTimetableVersionModal
- TimetableVersionCard

API 14: Archive Version

Endpoint

POST /api/principal/timetable/version/archive

Triggered By

- confirming ArchiveTimetableVersionModal

Request Body

```
{  
  "version_id": "TV_001",  
  "reason": "Superseded by newer timetable version"  
}
```

Response

```
{  
  "success": true,  
  "message": "Timetable version archived successfully",  
  "data": {
```

```
"version_id": "TV_001",  
  
"status": "archived"  
  
},  
  
"meta": {},  
  
"errors": []  
  
}
```

Used By Components

- ArchiveTimetableVersionModal
- TimetableVersionCard

API 15: Restore Version as Base Draft

Endpoint

POST /api/principal/timetable/version/restore-as-base

Triggered By

- clicking restore action on a version card

Request Body

```
{  
  
  "version_id": "TV_000"  
  
}
```

Response

```
{  
  
  "success": true,  
  
  "message": "Base draft created successfully from archived/published version",  
  
  "data": {  
  
    "new_draft_version_id": "TV_004",  
  
    "source_version_id": "TV_000"  
  
  },  
  
}
```

```
"meta": {},  
  
"errors": []  
  
}
```

Used By Components

- TimetableVersionCard

8) Session-Level Action APIs

API 16: Lock Timetable Entry

Endpoint

POST /api/principal/timetable/entry/lock

Triggered By

- TimetableSessionDetailsDrawer.onLock

Request Body

```
{  
  
  "entry_id": "TE_001"  
  
}
```

Response

```
{  
  
  "success": true,  
  
  "message": "Timetable entry locked successfully",  
  
  "data": {  
  
    "entry_id": "TE_001",  
  
    "is_locked": true,  
  
    "status": "locked"  
  
  },  
  
  "meta": {},  
  
}
```

```
"errors": []
```

```
}
```

API 17: Unlock Timetable Entry

Endpoint

POST /api/principal/timetable/entry/unlock

Request Body

```
{
```

```
  "entry_id": "TE_001"
```

```
}
```

Response

```
{
```

```
  "success": true,
```

```
  "message": "Timetable entry unlocked successfully",
```

```
  "data": {
```

```
    "entry_id": "TE_001",
```

```
    "is_locked": false,
```

```
    "status": "scheduled"
```

```
  },
```

```
  "meta": {},
```

```
  "errors": []
```

```
}
```

Used By Components

- TimetableSessionDetailsDrawer

API 18: Regenerate Related to Session

Endpoint

POST /api/principal/timetable/entry/regenerate-related

Triggered By

- regenerate action inside session details

Request Body

```
{  
  "entry_id": "TE_001",  
  "strategy": "local_repair"  
}
```

Response

```
{  
  "success": true,  
  "message": "Related timetable area regenerated successfully",  
  "data": {  
    "generation_run_id": "RUN_005",  
    "version_id": "TV_005",  
    "affected_entries_count": 3  
  },  
  "meta": {},  
  "errors": []  
}
```

Used By Components

- TimetableSessionDetailsDrawer

9) Component-to-API Mapping

PrincipalTimetablePage

Calls:

- GET /api/principal/timetable/summary
- GET /api/principal/timetable/readiness
- GET /api/principal/timetable/versions
- GET /api/principal/timetable/filter-options
- GET /api/principal/timetable/grid
- GET /api/principal/timetable/insights
- GET /api/principal/timetable/issues

TimetablePageHeader

Consumes:

- summary response only

TimetableStatusBanner

Consumes:

- summary response
- generation-status polling response

TimetableActionBar

Triggers:

- open modal only
- No direct API unless wired at container level

AltTimetableGenerationModal

Calls:

- POST /api/principal/timetable/generate
- then page polls GET /api/principal/timetable/generation-status

PartialRegenerationModal

Calls:

- POST /api/principal/timetable/regenerate-partial

TimetableViewControls

Triggers grid reload via:

- GET /api/principal/timetable/grid

TimetableGrid

Consumes:

- grid API response

TimetableCell

Triggers:

- GET /api/principal/timetable/session-details

TimetableInsightsPanel

Consumes:

- GET /api/principal/timetable/insights

TimetableVersionManager

Consumes:

- GET /api/principal/timetable/versions

Triggers:

- publish
- archive
- restore

TimetableIssuesSection

Consumes:

- GET /api/principal/timetable/issues

TimetableDiagnosticsModal

Consumes:

- GET /api/principal/timetable/diagnostics

TimetableSessionDetailsDrawer

Consumes:

- GET /api/principal/timetable/session-details

Triggers:

- lock
- unlock

- regenerate related

10) Recommended Frontend Service Layer

موحدًا هكذا service file يفضل أن تنشئ:

```
// timetableApi.ts

export async function fetchTimetableSummary(params)

export async function fetchTimetableReadiness(params)

export async function fetchTimetableVersions(params)

export async function fetchTimetableFilterOptions(params)

export async function fetchTimetableGrid(params)

export async function fetchTimetableInsights(params)

export async function fetchTimetableIssues(params)

export async function fetchTimetableDiagnostics(params)

export async function fetchTimetableSessionDetails(params)


export async function generateFullTimetable(payload)

export async function fetchGenerationStatus(params)

export async function partiallyRegenerateTimetable(payload)


export async function publishTimetableVersion(payload)

export async function archiveTimetableVersion(payload)

export async function restoreTimetableVersionAsBase(payload)


export async function lockTimetableEntry(payload)

export async function unlockTimetableEntry(payload)

export async function regenerateRelatedEntry(payload)
```

11) Suggested Error Codes

موحدة مثل error codes يعيد backend اجعل ال:

READINESS_BLOCKED

INVALID_CONTEXT

VERSION_NOT_FOUND

ENTRY_NOT_FOUND

BLOCKING_CONFLICTS

GENERATION_ALREADY_RUNNING

INSUFFICIENT_INPUT_DATA

TEACHER_MAPPING_MISSING

CLASS_MAPPING_INVALID

UNAUTHORIZED_ACTION

LOCKED_ENTRY_CANNOT_BE_MODIFIED

12) UX Behavior After Each API

After generate

- close modal
- show processing banner
- start polling generation status
- disable publish/archive until completion

After generation-status = completed

- stop polling
- refresh:
 - summary
 - versions
 - grid
 - insights
 - issues
- show success toast

After publish

- refresh summary + versions + grid
- show success banner/toast

After archive

- refresh versions
- if archived version was selected, switch to valid fallback version

After lock/unlock entry

- refresh session details + grid
- show success toast

13) Short Handoff Version for Vibe Coding Agent

Build a REST API contract for the Principal Timetable Page.

Core APIs:

- GET /api/principal/timetable/summary
- GET /api/principal/timetable/readiness
- GET /api/principal/timetable/versions
- GET /api/principal/timetable/filter-options
- GET /api/principal/timetable/grid
- GET /api/principal/timetable/insights
- GET /api/principal/timetable/issues
- GET /api/principal/timetable/diagnostics
- GET /api/principal/timetable/session-details

Mutation APIs:

- POST /api/principal/timetable/generate
- GET /api/principal/timetable/generation-status
- POST /api/principal/timetable/regenerate-partial
- POST /api/principal/timetable/version/publish
- POST /api/principal/timetable/version/archive
- POST /api/principal/timetable/version/restore-as-base

- POST /api/principal/timetable/entry/lock
- POST /api/principal/timetable/entry/unlock
- POST /api/principal/timetable/entry/regenerate-related

Each endpoint must return typed, structured responses with success, message, data, meta, and errors.